



**Technical
Short Course**



Strain Accumulation & Release from GNSS: Earth Structure Effects

June 17, 2026 | 9:00 am – 1:00 pm PT

Instructors

Led by CRESCENT Coupling, Seismicity, & Slow Slip (C3S) Working Group

Jack Loveless
Smith College



Brendan Crowell
The Ohio State U.



Kaj Johnson
Indiana U. Bloomington



Tim Melbourne
Central Washington U.



Shannon Fasola

CRESCENT Geoscience Education & Inclusion Program Manager
(Made it happen)

Thank You



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GEOPHYSICAL
FACILITY**
Operated by EarthScope

GeoLab

Moodle Course Page

Sponsored Microcredential through Credly

Day 3 Overview



Time (PT)	Topic
9:00 AM	Earth Structure Effects and Slip Inversion Modeling with GeoSlip2D
10:00 AM	Layered Elastic Models
10:30 AM	10-minute Break
10:40 AM	Compliant Wedge Model
11:50 AM	10-minute Break
12:00 PM	Viscoelastic Earthquake Cycle Models
1:00 PM	Course adjourns for day Optional 30 min Q&A with instructors

Housekeeping



- **Slack Channel:** #2026_geodesy_tsc
 - Ask questions
 - Connect with other participants before, during, and after the course
 - Receive course updates and announcements
- **Moodle page:** courses.earthscope.org
 - Login with your EarthScope Account
- Lectures will be recorded and posted to Moodle each day

Coupling, Seismicity, and Slow Slip Working Group (C3S)



Q1: How does tectonics drive hazard?

Q2: How is strain accumulated and released throughout the EQ cycle?

Q3: What is the temporal and spatial variability of past earthquakes?

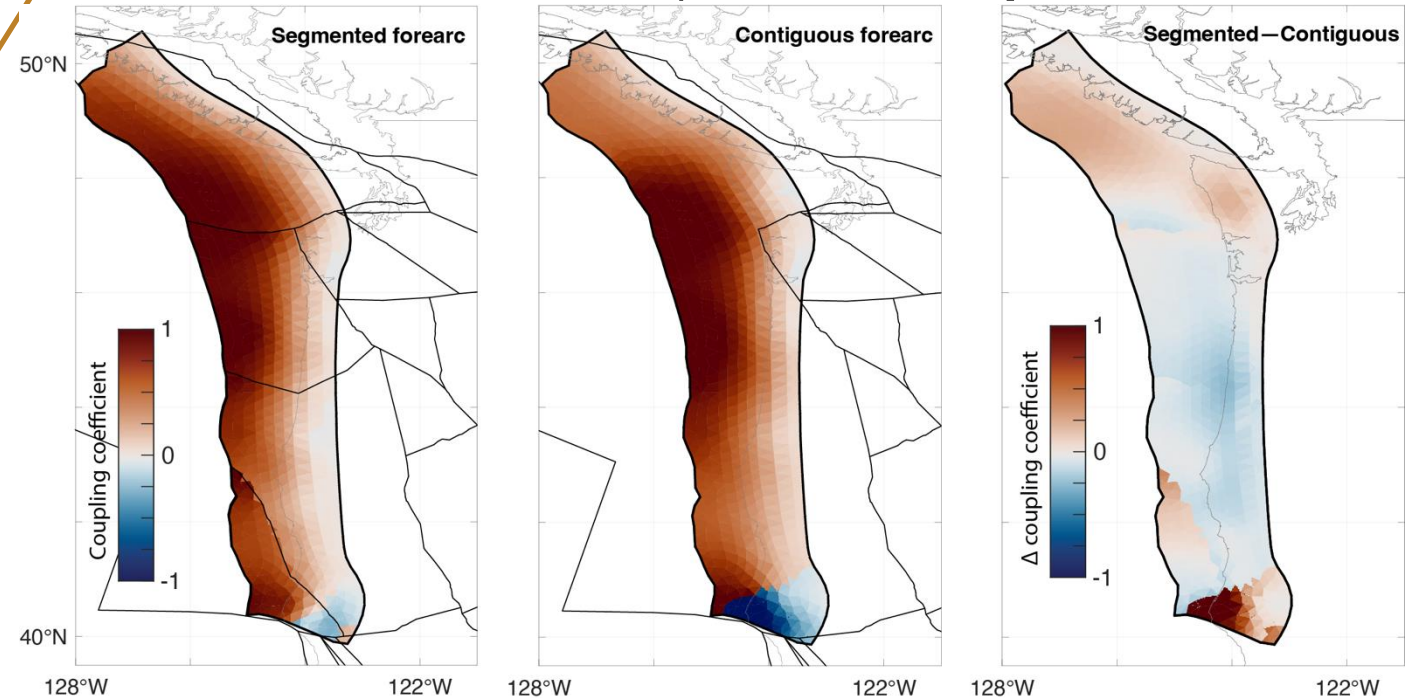
Q4: What are the controls on rupture dynamics during megathrust events?

Q5: What is the seismic hazard from intra-plate (crust and slab) earthquakes?

Q6: What is the nature of strong motion in the CSZ?

Q7: What are the tsunami sources in the CSZ and what controls inundation?

- **Lithospheric block models** to evaluate the strain budget of the PNW for seismic coupling and more
- **Seismogeodetic** detection of **transient** events
- New, richer inventory of **seismicity**



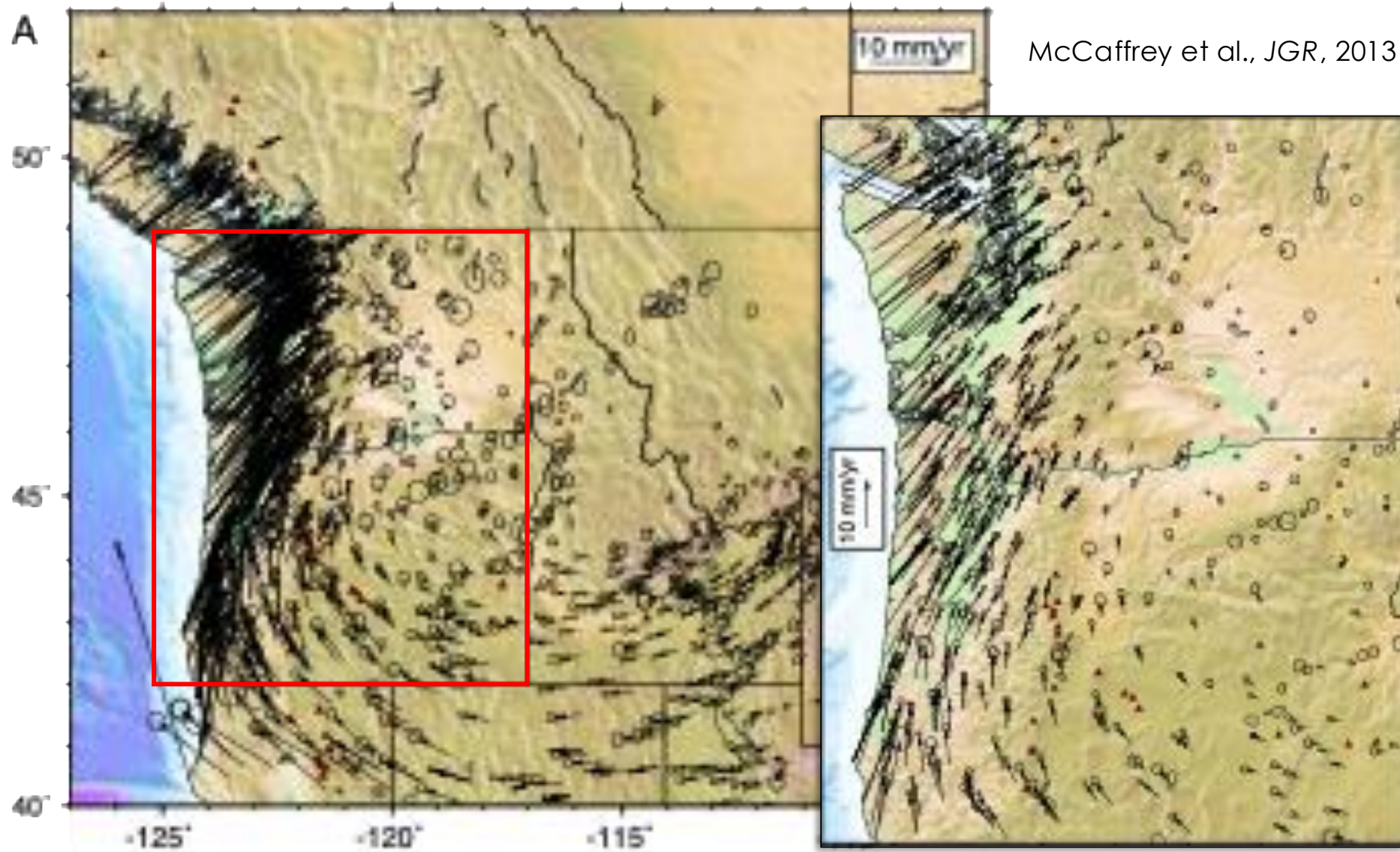
H. Elston, J. Loveless, et al. 2025

Interseismic block modeling



McCaffrey et al., *JGR*, 2013

In Cascadia, considering the effects of crustal block rotation plus strain accumulation related to subduction zone coupling is consistent with a first-order look at the velocity field



Interseismic block modeling

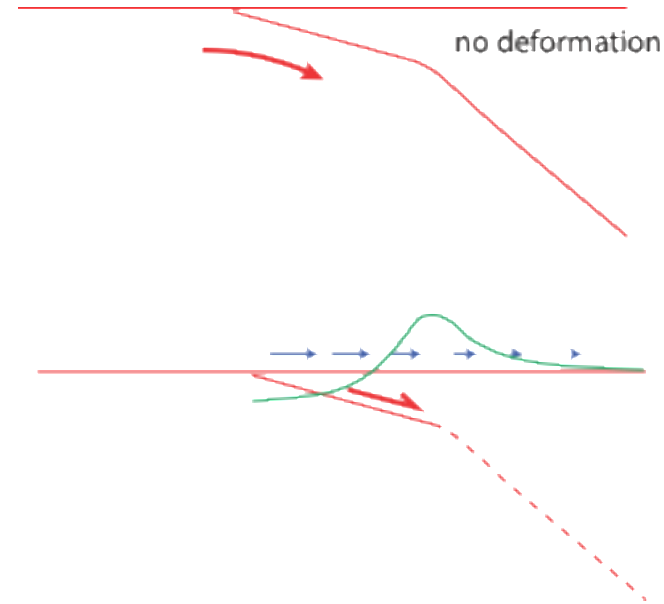
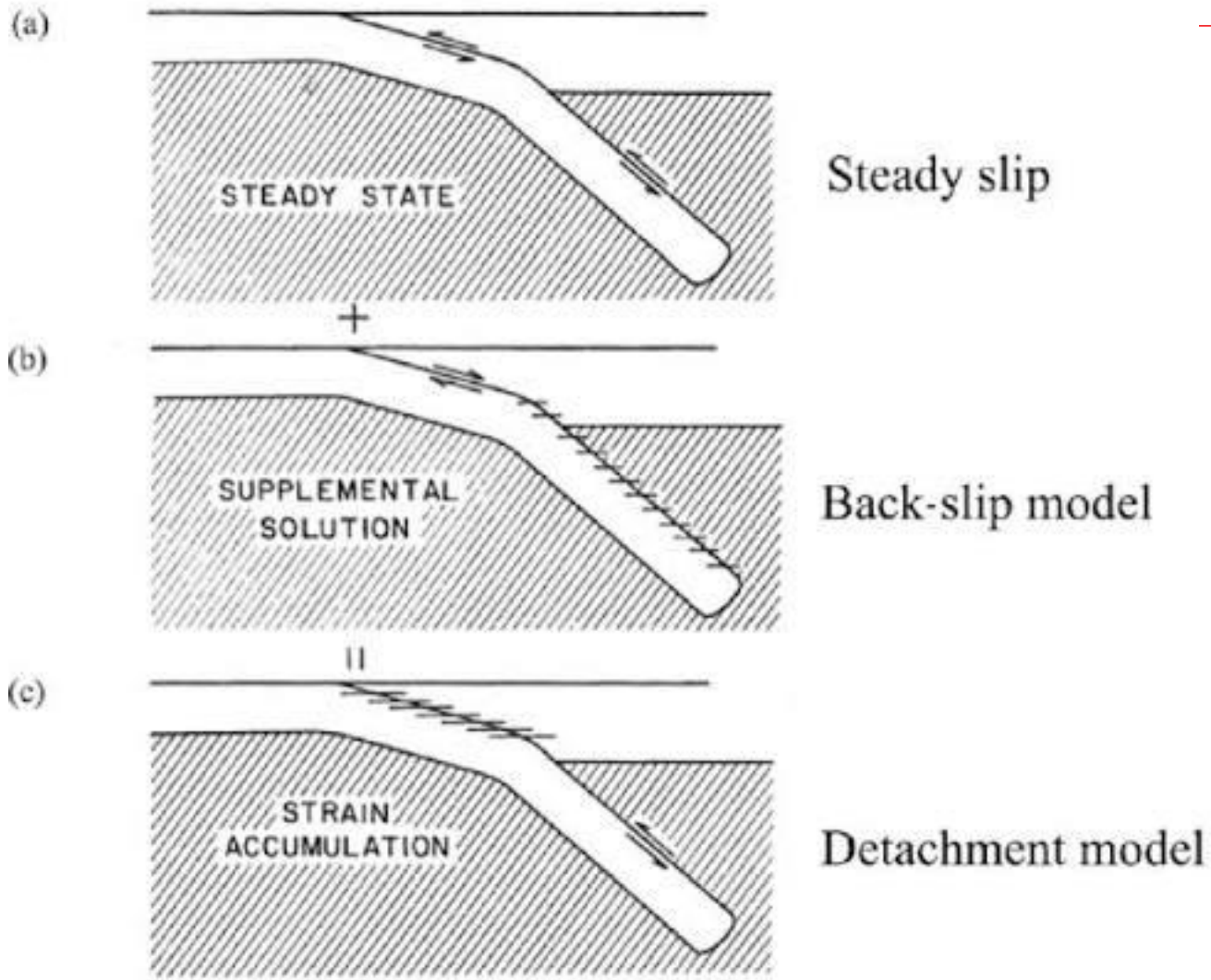


Decompose interseismic velocity field into contributions from:

- Block rotation,
- Elastic strain accumulation on block-bounding faults,
- Elastic strain accumulation on subduction zones,
- Intrablock strain and volcanic inflation

$$\underline{\mathbf{V_I}} = \underline{\mathbf{V_B}} - \underline{\mathbf{V_E}} - \underline{\mathbf{V_T}}$$

Interseismic Coupling at Subduction Zones



- Elastic halfspace
- Edge dislocation
- Plane strain
(infinitely long fault)

JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 88, NO. B6, PAGES 4984-4996, JUNE 10, 1983

A Dislocation Model of Strain Accumulation and Release at a Subduction Zone

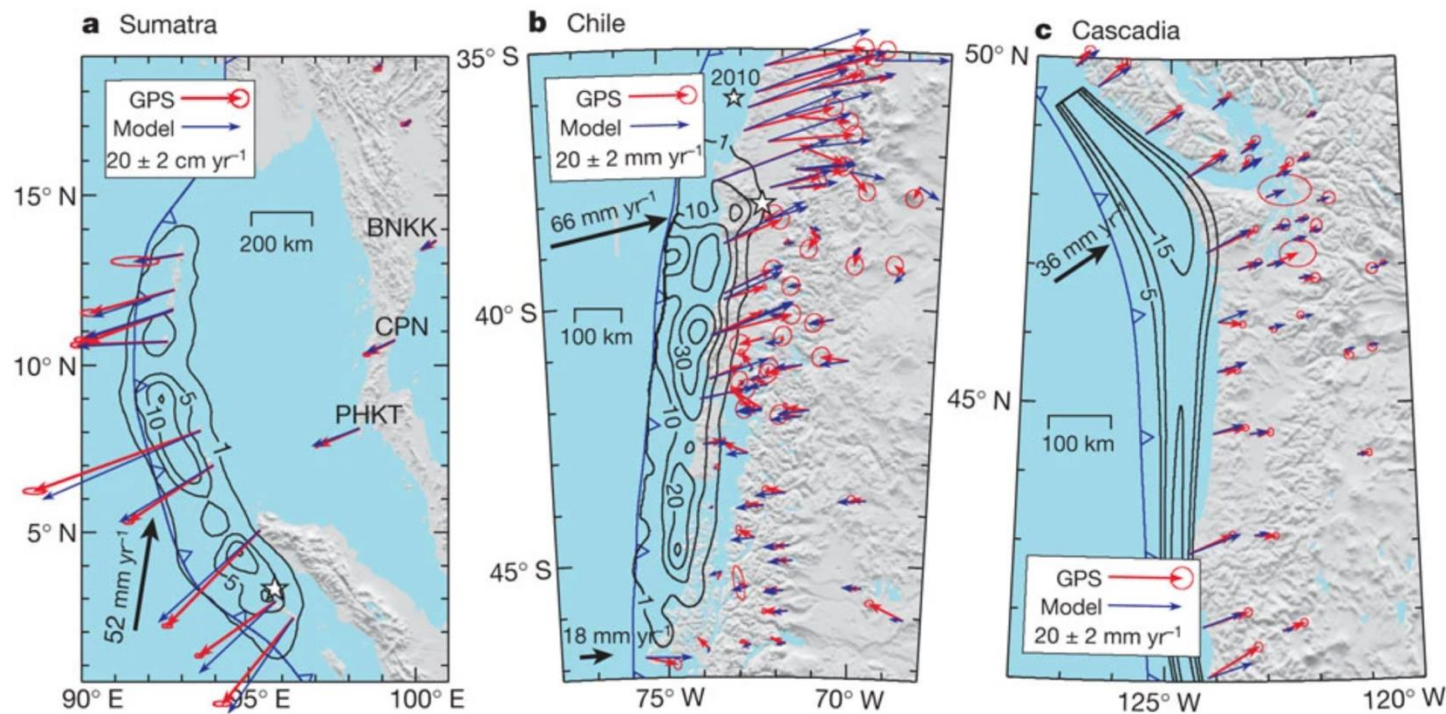
J. C. SAVAGE

U.S. Geological Survey, Menlo Park, California 94025

Earthquake Cycle

Figure 3: GPS- (red) and model-predicted (blue) surface velocities for three subduction zones that are at different stages of the earthquake cycle.

From: [Deformation cycles of subduction earthquakes in a viscoelastic Earth](#)



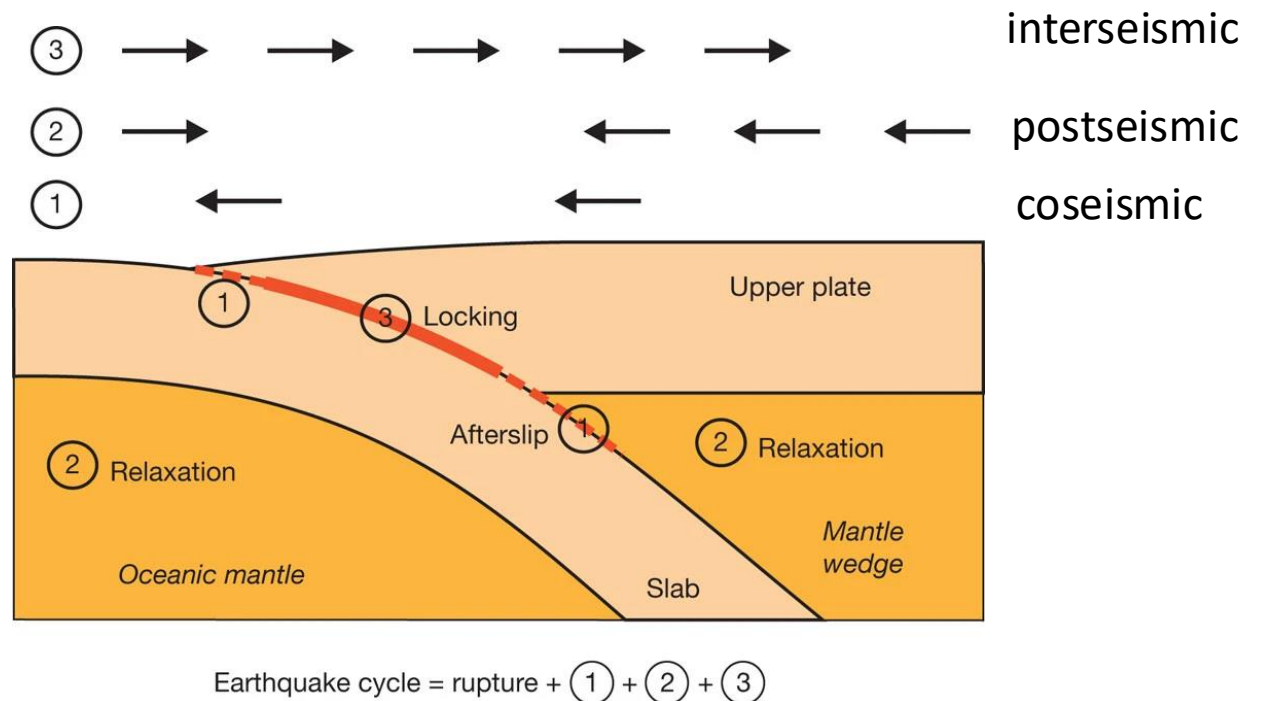
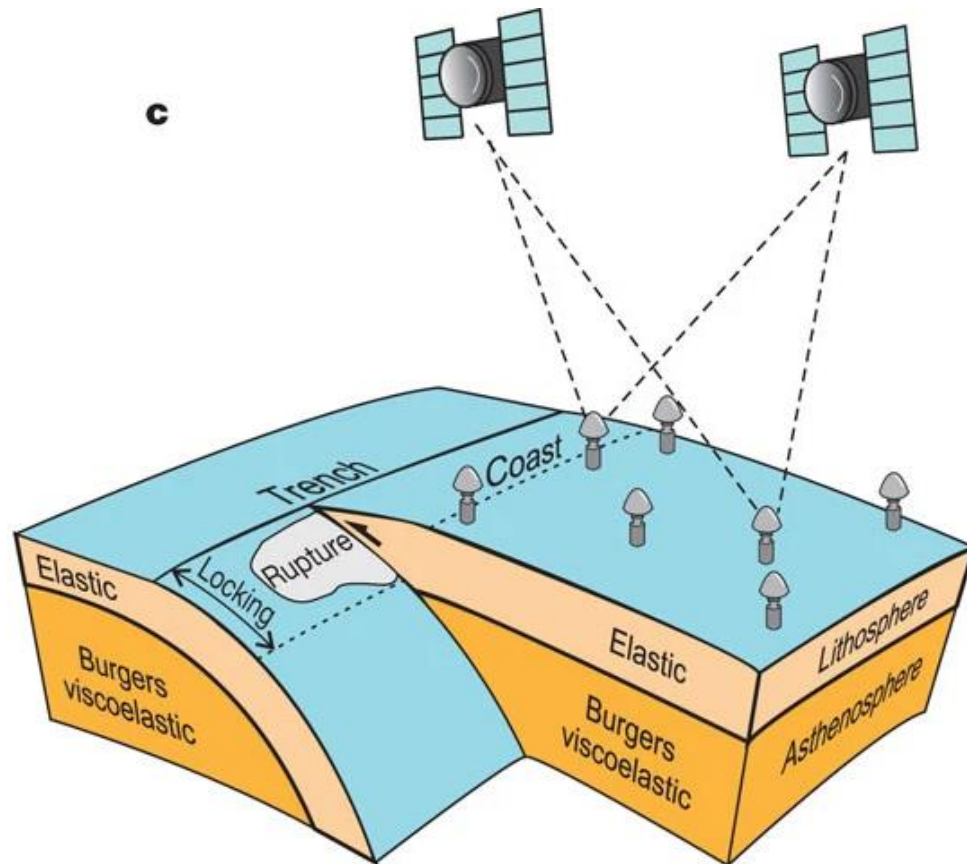
Early postseismic phase

Late postseismic phase

Late interseismic phase

Wang et al., 2012

Viscoelastic Deformation Earthquake Cycle

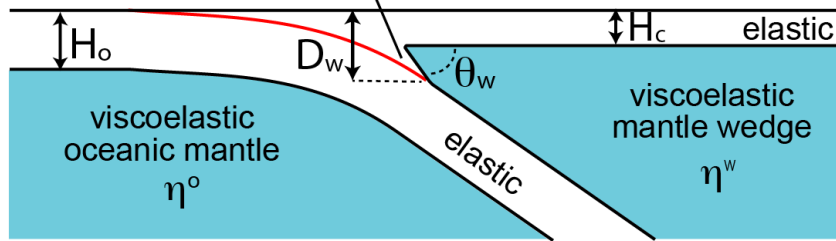


Wang et al., 2012

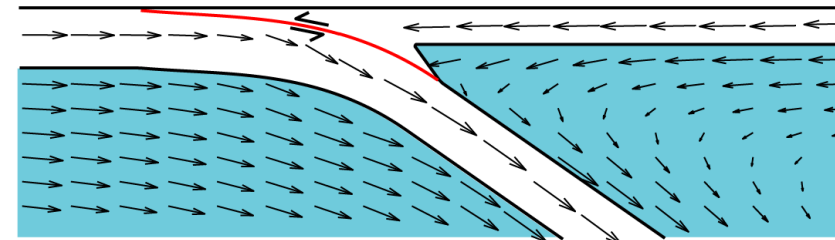
Viscoelastic Deformation Earthquake Cycle



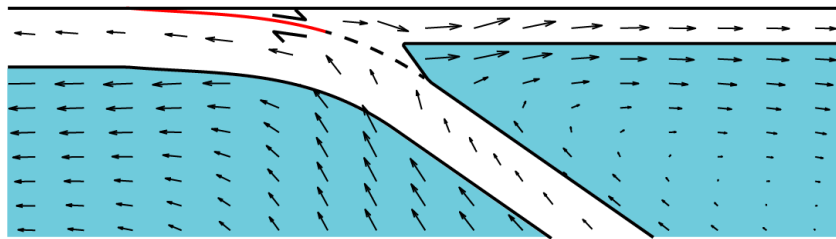
(a) Subduction zone geometry
cold wedge



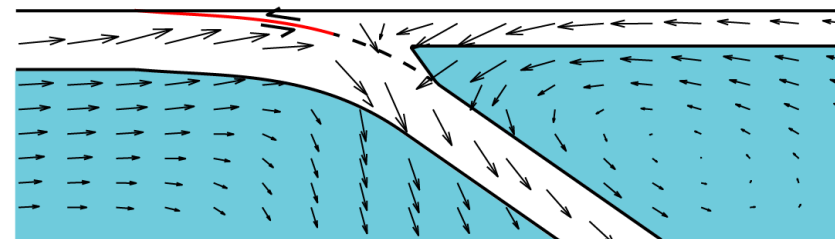
(b) long term (steady mantle flow)



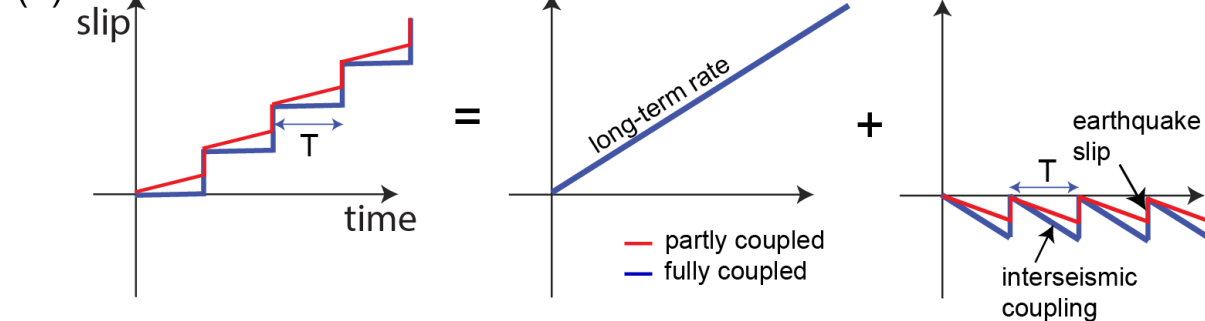
(c) interseismic coupling (steady flow)



(d) repeated earthquakes (transient flow)

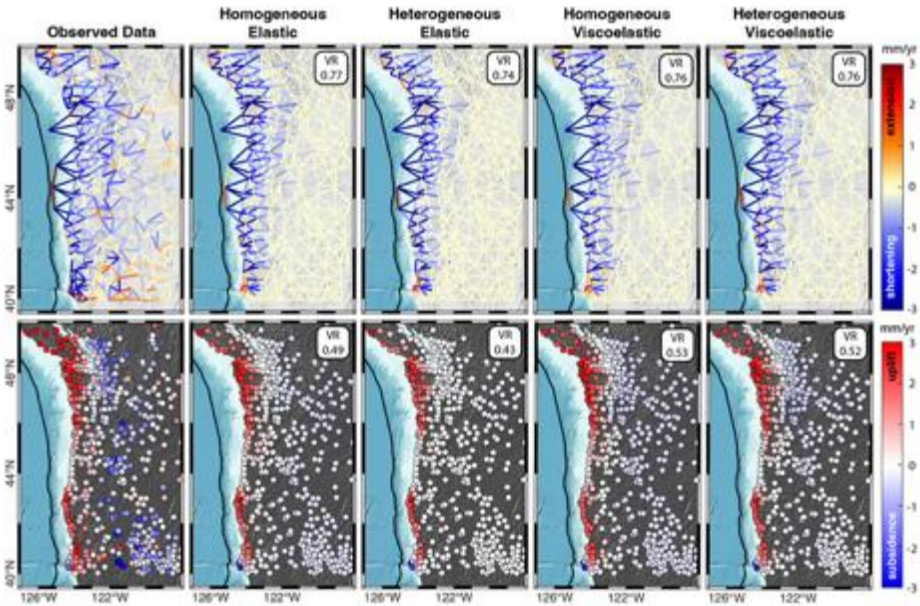


(e) periodic slip history

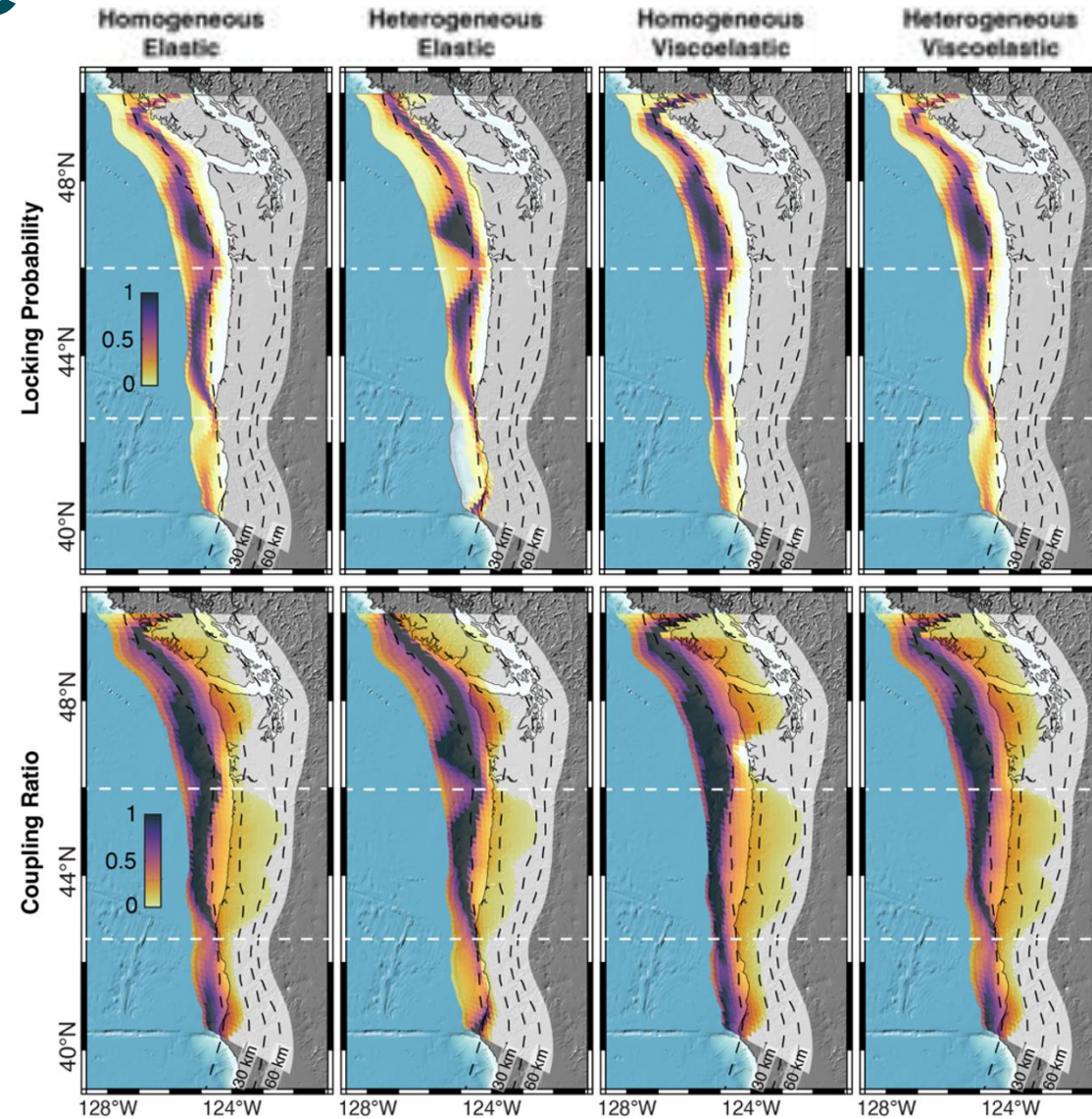


Sherrill et al., 2026 (in review)

Viscoelastic Deformation Earthquake Cycle

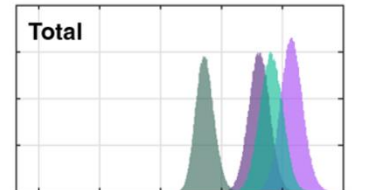
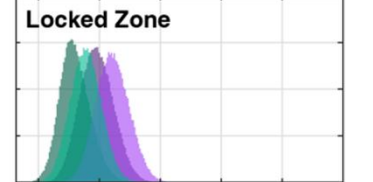


Sherrill et al., 2026 (in review)



b.

Moment Accumulation Rate
[$\times 10^{19}$] (N m/year)



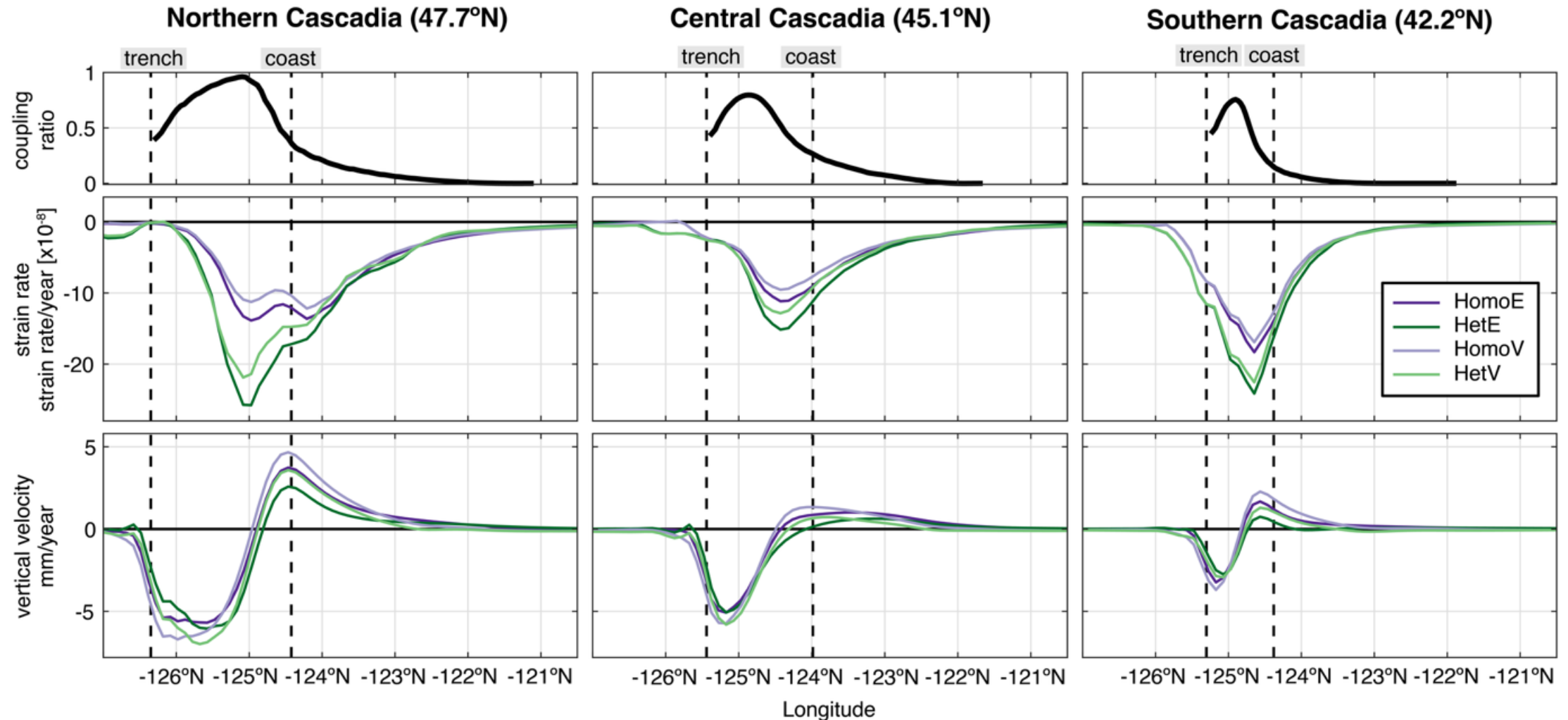
8.63 8.83 8.95 9.03 9.10
8.49 8.69 8.80 8.89 8.95
Magnitude
(300-year & 500-year cycle)

Homo. Elastic, Het. Elastic, Homo. Visco., Het. Visco.

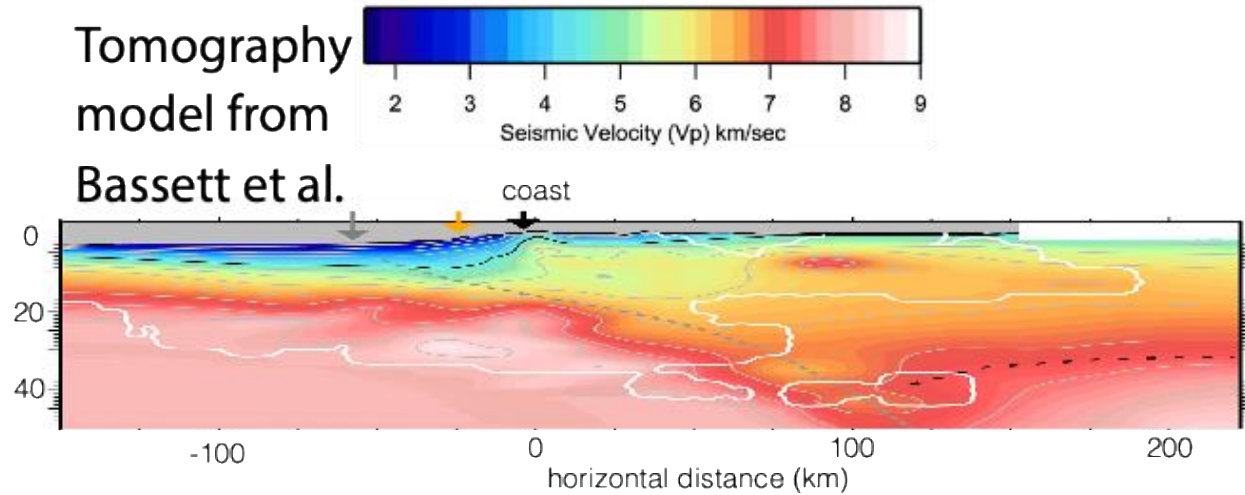
Viscoelastic Deformation Earthquake Cycle



Sherrill et al., 2026 (in review)



Influence of Elastic Properties

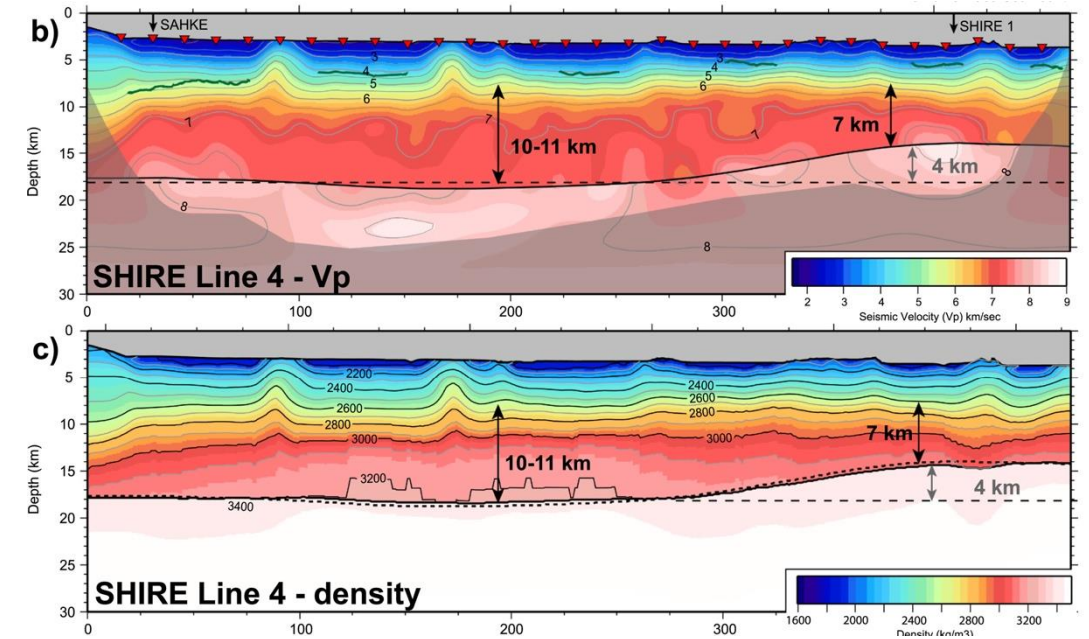


For an isotropic elastic medium, seismic velocities are related to elastic moduli by:

$$V_p = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho}}, \quad V_s = \sqrt{\frac{\mu}{\rho}}$$

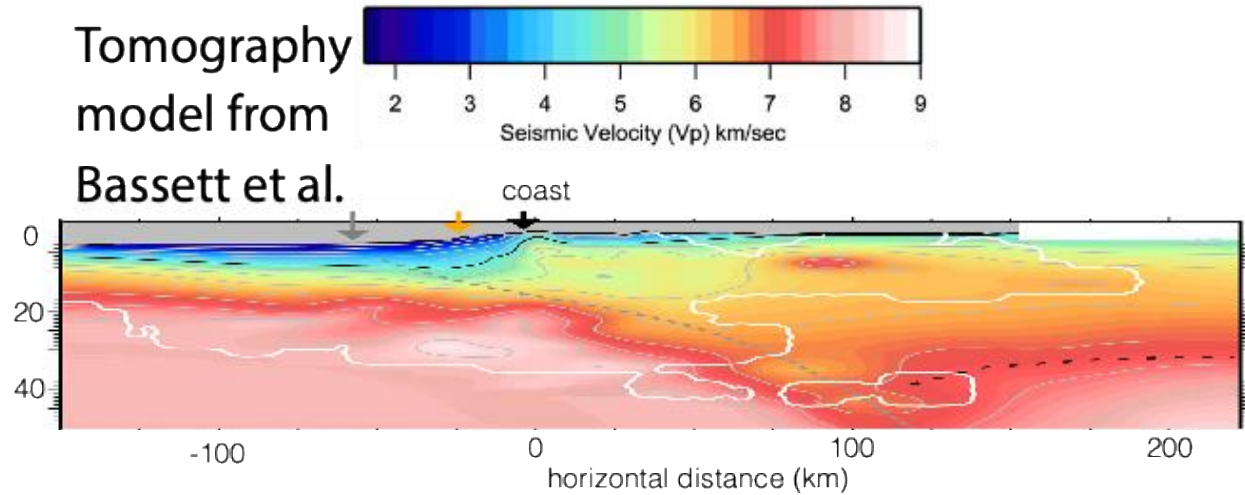
where:

- V_p = P-wave velocity
- V_s = S-wave velocity
- ρ = density
- K = bulk modulus
- μ = shear modulus, also called rigidity
- λ = Lamé's first parameter



Bassett et al., 2022, 2023

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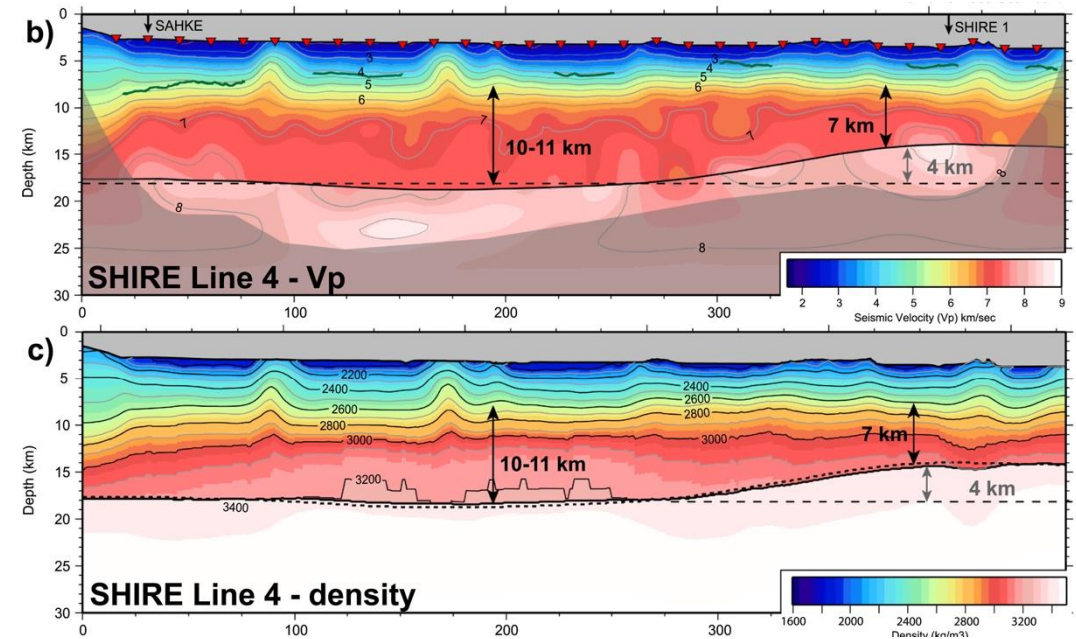
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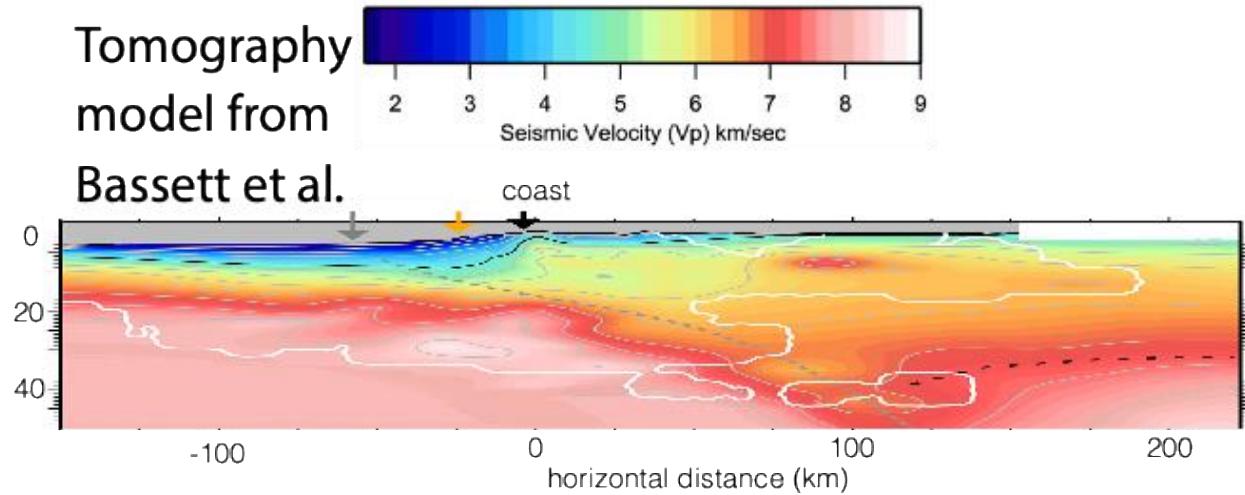
$$\mu = \rho V_p^2 \frac{1 - 2\nu}{2(1 - \nu)}$$

For rocks, assuming $\nu \approx 0.25$, this simplifies to:

$$\mu \approx \frac{1}{3} \rho V_p^2$$



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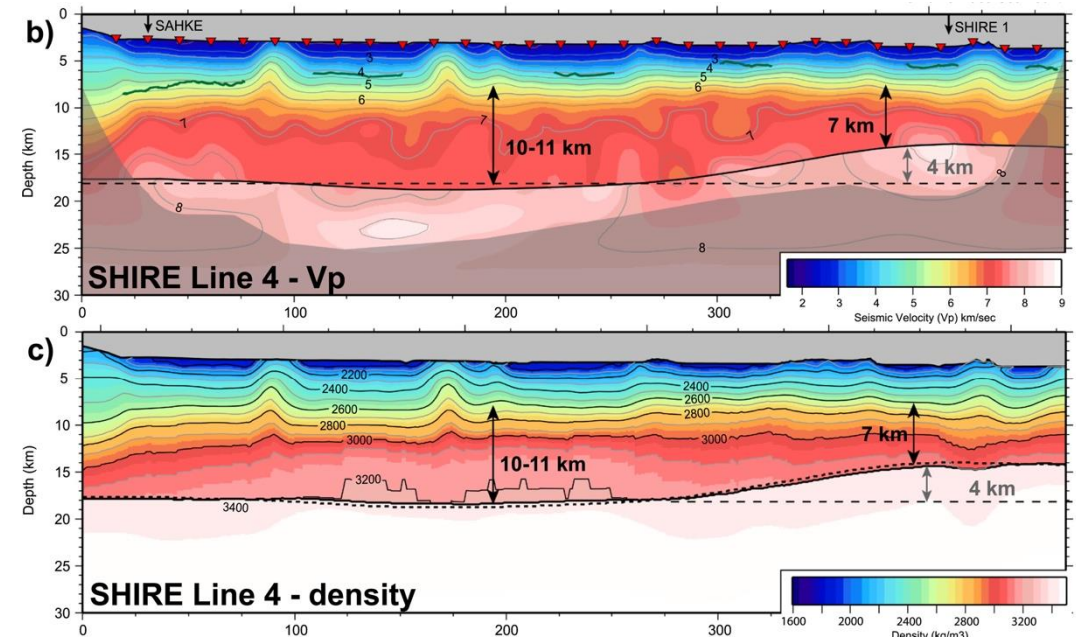
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$$\frac{\mu_{\text{shallow}}}{\mu_{\text{deep}}} = \frac{1800(2)^2}{3000(6)^2}$$

$$\frac{\mu_{\text{shallow}}}{\mu_{\text{deep}}} \approx 0.067$$

$$\mu_{\text{deep}} \approx 15 \mu_{\text{shallow}}$$

GeoSlip2D

2D plane strain modeling package in Python



GeoSlip2D

GeoSlip2D is a Python package for computing two-dimensional Green's functions for slip on a subduction-interface profile and using those Green's functions in 1D profile slip inversions.

The package is designed around a common workflow:

```
interface geometry
  ↓
Green's-function backend
  ↓
Greens2D object
  ↓
save / load
  ↓
profile slip inversion
```

GeoSlip2D currently includes Green's-function tools for:

- homogeneous elastic half-space calculations,
- layered elastic models,
- compliant elastic wedge models,
- viscoelastic earthquake-cycle models,
- and profile slip inversions using horizontal and optionally vertical surface velocities.

The package is intended for testing how assumptions about elastic structure and earthquake-cycle processes affect inferred slip or slip-deficit distributions on a 2D dipping fault interface.

Platform ▾ Solutions ▾ Resources ▾ Open Source ▾ Enterprise ▾ Pricing

[kajjohns](#) / [GeoSlip2D](#) Public

<> Code Issues Pull requests Actions Projects Security and quality Insights

main 1 Branch 0 Tags <> Code ▾

kajjohns	Restructure repository layout and update README	bf5b270 · last week	🕒 2 Commits
📁 data	Restructure repository layout and update README	last week	
📁 geoslip2d	Restructure repository layout and update README	last week	
📁 images	initial commit	3 weeks ago	
📄 .gitignore	initial commit	3 weeks ago	
📄 Compliant_wedge_model_demo.ipynb	Restructure repository layout and update README	last week	
📄 Layered_model_demo.ipynb	Restructure repository layout and update README	last week	
📄 README.md	Restructure repository layout and update README	last week	
📄 Slip_inversion_activities.ipynb	Restructure repository layout and update README	last week	
📄 Slip_inversion_demo.ipynb	Restructure repository layout and update README	last week	
📄 Viscoelastic_cycle_demo.ipynb	Restructure repository layout and update README	last week	
📄 pyproject.toml	initial commit	3 weeks ago	

Launching GeoLab & Accessing Course GitHub Repository



If you **have** completed pre-course tutorials, please do the following now:

- In Moodle, go to **Day 3** section
 - Click the permalink to launch **GeoLab (make sure you have 29 GB selected)**
- Follow the steps in the GitHub Pre-Course Tutorial (Part 2) to:
 - **Sync your repo to receive most recent updates from instructor course repo**

A screenshot of the JupyterHub 'Server Options' interface. The header shows the JupyterHub logo and navigation links for Home, Token, and Usage. On the right, it displays a user token 'auth0|67d0ae0cbf1784f4c7b1e67c' and a 'Logout' button. The main section is titled 'Server Options' and contains a box for 'Choose your environment and resources' with a 'Permalink' link. Inside this box, there are two dropdown menus: 'Environment' set to 'GeoLab' and 'Resource Allocation' set to '29 GB RAM, ~4 CPUs' with a subtext '~4 CPUs always available'. At the bottom of the box is an orange 'Start' button.

Schedule of Notebook Activities



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 [Slip_inversion_demo.ipynb](#)

 [Layered_model_demo.ipynb](#)

 [Compliant_wedge_model_demo.ipynb](#)

 [Viscoelastic_cycle_demo.ipynb](#)

 [Slip_inversion_activities.ipynb](#)

Independent Exercise 1



 [Slip_inversion_demo.ipynb](#)

1. In Section 6, experiment with different slip inversion configurations.
 - (a) Vary alpha systematically
 - (b) Change to “bounded” inversion and experiment with upper and lower bounds
 - (c) Play around with other parameters

2. Go back to Section 1 and change the data set to "synthetic_slowslip_data". In Section 6, change the inversion_mode to "forward_slip". Now rerun all the cells to conduct the inversion

Independent Exercise 2



 Layered_model_demo.ipynb

Experiment with different layering and explore the influence on surface deformation

Vary Layer depths, h , (note, don't specify depths below about 40 km – solution can be unstable)
Vary relative shear moduli, μ

```
h=np.array([5.0, 10.0, 15.0]),  
mu=np.array([0.1, 3.0, 5.0, 10.0]),
```

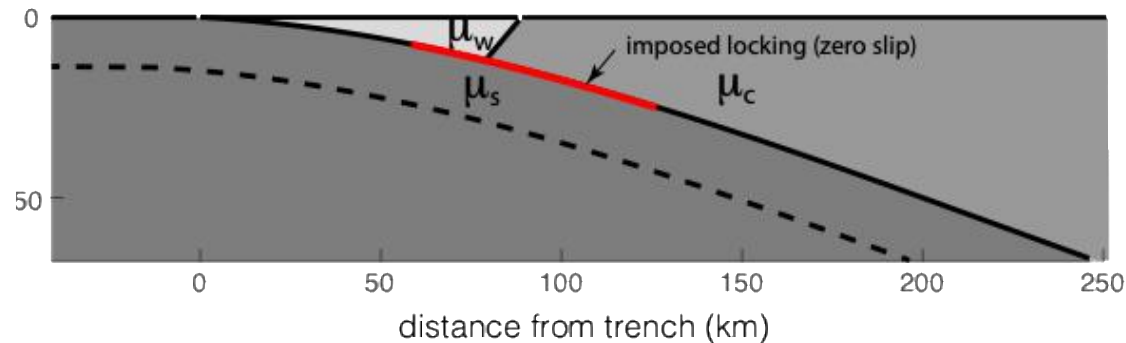
Independent Exercise 3



Compliant_wedge_model_demo.ipynb

Experiment with different relative elastic moduli for the wedge model

$\mu_1=1.0$,
 $\mu_2=0.1$, # compliant wedge
 $\mu_3=10.0$,



Note:

$\mu_1 = \mu_c$

$\mu_2 = \mu_w$

$\mu_3 = \mu_w$

Independent Exercise 4



 Viscoelastic_cycle_demo.ipynb

In the # FAST STEP of Section 2 (Build and refresh Green's functions), modify the relaxation time and earthquake timing parameters and run subsequent cells to see what happens.

```
forward_tR = 10.0 # relaxation time scale  
forward_teq = 500.0 # time since last earthquake (years)  
forward_T = 550.0 # earthquake recurrence time (years)
```

Independent Exercise 5



 Slip_inversion_activities.ipynb

Run this notebook to conduct 1D profile slip inversions of your data set(s) generated on Day 1 of this course.

Try inversions of the real data set using different Earth models.



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**Technical
Short Course**



Strain Accumulation & Release from GNSS

Wrap Up



Access to Course Materials & Continued Support



- You will continue to have access to the **Moodle course page** indefinitely.
- **GeoLab** is available to those who have an EarthScope account.
 - Remember: GeoLab shouldn't be used for long-term storage.
- Be sure to push any changes from your local repo on GeoLab to your forked repo on your personal GitHub account.
 - Instructions available in GitHub tutorial on Moodle.
- Course **requirements file** is available in the course repo.
- Continue using the **course Slack channel (#2026_geodesy_tsc)** to stay connected with each other, ask questions, share resources, and help each other as you continue applying these skills.

Course Feedback Survey



- You will receive an email from **Michael Coe** (qualtricssurvey@uoregon.edu) containing a link to an anonymous **Qualtrics** survey.
- The survey should take only a few minutes to complete.
- Your feedback is important to us and helps us improve future technical short courses.

Thank you for your participation and for sharing your feedback!

Awareness-Level Microcredential through Credly



- Participants who attend **all 3 days** of the short course will be awarded an **Awareness-Level Credly Microcredential**.
- **What it means:**
 - This digital badge recognizes your **participation** and **engagement** in the course activities — a valuable addition to your professional profile or LinkedIn!
- You'll receive your microcredential **via email after the course**.



Contact Us:

cascadiaquakes@uoregon.edu



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